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POTENTIAL FLOW

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NOMENCLATURE

a	Cylinder radius	m
C_p	Pressure coefficient	$-$
D', L'	Drag and lift per unit span	N/m
F	Complex potential	m^2/s
m	Source strength (volume flow rate per unit depth)	m^2/s
n	Sector exponent, $n = \pi/\beta$	$-$
r, θ	Polar coordinates	m, °
u, v	Velocity components in x and y	m/s
V_r, V_θ	Radial and tangential velocity	m/s
V_∞	Free-stream speed	m/s
V	Sector flow strength constant	$\mathrm{m}^{2-\mathrm{n}}/\mathrm{s}$
w	Complex velocity, $w = u - iv$	m/s
x_0, y_0	Position of an element	m
z	Complex coordinate, $z = x + iy$	m
α	Direction of the uniform flow	°
β	Opening angle of the sector	°
Γ	Circulation (positive counter-clockwise)	m^2/s
δ	Axis angle of the doublet	°
θ_1, θ_2	Wall angles of the sector	°
θ_w	Direction angle of a symmetry wall	°
μ	Doublet strength	m^3/s
ρ_∞	Free-stream density	kg/m^3
ϕ, ψ	Velocity potential, stream function	m^2/s

1. INTRODUCTION

A two-dimensional, incompressible and irrotational flow can be built by adding elementary flows, because its governing equation is linear. In this project, a graphical program (Flowy) was written in Python for this purpose. Uniform flow, sources and sinks, point vortices, sector (wedge) flow, doublets and straight symmetry walls are selected by the user, and the streamlines with direction arrows are recomputed and redrawn after every change. The program can be run on Windows 10 and 11 without a Python installation.

The equations used by the program are given in Section 2, the program architecture in Section 3, the validation against analytic solutions in Section 4 and worked examples in Section 5.

2. GOVERNING EQUATIONS

2.1 Potential Flow and the Complex Potential

For an incompressible and irrotational flow, Laplace's equation is satisfied by the velocity potential ϕ ([1], Eq. (3.40)):

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0 \tag{1}$$

It is also satisfied by the stream function ψ, and the streamlines are given by $\psi = \text{const}$. With $z = x + iy = re^{i\theta}$, the complex potential and the complex velocity are ([2], Sec. 7.4):

$$F(z) = \phi + i\psi, \qquad w(z) = \frac{dF}{dz} = u - iv = (V_r - iV_\theta)e^{-i\theta} \tag{2}$$

Since Eq. (1) is linear, the potentials and velocities of N elements are added. In the program, the velocities are added at every point and the streamlines are integrated in the total field; pictures of streamlines are never added:

$$F(z) = \sum_{k=1}^{N} F_k\,(z), \qquad w(z) = \sum_{k=1}^{N} w_k\,(z) \tag{3}$$

2.2 Elementary Flows

The elementary flows and their complex potentials are given in Table 1, where $z_0 = x_0 + iy_0$ is the position of an element. The circulation Γ is positive counter-clockwise; in [1] the opposite sign is used, so $\Gamma_{cw} = -\Gamma$ is used with the equations of [1]. In the notation of [1], $m = \Lambda$ and $\mu = \kappa/2\pi$.

Table 1. Elementary flows: inputs and complex potentials

Element	Inputs	Complex potential $F(z)$	Source
Uniform flow	speed V_∞, angle α	$V_\infty e^{-i\alpha} z$	[1], Eqs. (3.53), (3.55)
Source / sink	m (source if $m > 0$), x_0, y_0	$(m/2\pi)\ln(z - z_0)$	[1], Eqs. (3.67), (3.72)
Point vortex	Γ (ccw if $\Gamma > 0$), x_0, y_0	$-i(\Gamma/2\pi)\ln(z - z_0)$	[1], Eqs. (3.112), (3.114)
Doublet	μ, x_0, y_0, axis angle δ	$\mu e^{i\delta}/(z - z_0)$	[1], Eqs. (3.87), (3.88)
Sector flow	V, x_0, y_0, θ_1, θ_2	$V\left[(z - z_0)e^{-i\theta_1}\right]^n$, $n = \pi/\beta$	[2], Sec. 7.4
Wall	x_w, y_w, θ_w, fluid side	images, Eq. (5)	[3]

For the sector flow with the opening angle $\beta = \theta_2 - \theta_1$, the stream function is

$$\psi = V r^n \sin[n(\theta - \theta_1)], \qquad n = \frac{\pi}{\beta} \tag{4}$$

where r and θ are measured from the vertex. Since $\psi = 0$ at $\theta = \theta_1$ and at $\theta = \theta_2$, both walls are streamlines.

2.3 Symmetry Walls and the Method of Images

A symmetry wall passes through $z_w = x_w + iy_w$ with the angle θ_w. With the mirror point z_m of z, the total complex potential is ([3]):

$$z_m = z_w + e^{2i\theta_w}\,\overline{(z - z_w)}, \qquad F_{total}(z) = F(z) + \overline{F(z_m)} \tag{5}$$

By Eq. (5), the wall is made a streamline, so $V \cdot n = 0$ is satisfied exactly. A source is mirrored as a source of the same strength, a vortex as a vortex of opposite sign and a doublet with a mirrored axis. For two walls at $\beta = 180°/k$, $2k - 1$ images are needed; for two parallel walls, the infinite row of images is summed in closed form, $\sum_n 1/(z - a - nP) = (\pi/P)\cot[\pi(z - a)/P]$. No image is used for a uniform flow, so it must be parallel to every wall.

2.4 Admissible Domain of the Sector Flow

Unrestricted superposition does not keep the walls of a sector flow. Therefore the following rules are used in the program. (i) The fluid region of a sector flow is only the wedge $\theta_1 \le \theta \le \theta_2$. (ii) Every other element must lie inside the wedge; an element on a wall or outside the wedge is reported as an error. (iii) The two sides of the wedge are treated as symmetry walls for all other elements, so the images of Section 2.3 are added for these elements. This is exact only for $\beta = 180°/k$ (e.g. 90°, 60°, 45°); for other angles an image would fall inside the fluid, and the scene is reported as an error. (iv) For $\beta > 180°$, no other element can be added, because the walls cannot be represented by images. (v) Only one sector flow is allowed in a scene, and it is not combined with symmetry walls, because it already has its own walls. (vi) The sector flow itself is not imaged, because Eq. (4) already satisfies its walls.

2.5 Pressure Coefficient and Forces

The pressure coefficient is obtained from Bernoulli's equation ([1], Eq. (3.38)):

$$C_p = 1 - \left(\frac{V}{V_\infty}\right)^2 \tag{6}$$

A cylinder of radius $a = \sqrt{\mu/V_\infty}$ is obtained from a uniform flow and a doublet with $\delta = \alpha$ ([1], Eq. (3.98)). With a vortex Γ at its center, the surface velocity and the stagnation points are ([1], Eqs. (3.125) and (3.123)), and the lift and drag per unit span are ([1], Eqs. (3.140) and (3.132)):

$$V_\theta = -2V_\infty \sin\theta + \frac{\Gamma}{2\pi a}, \qquad \sin\theta_s = \frac{\Gamma}{4\pi V_\infty a} \tag{7}$$

$$L' = \rho_\infty V_\infty \Gamma_{cw} = -\rho_\infty V_\infty \Gamma, \qquad D' = 0 \tag{8}$$

In the program, the forces are computed by integrating the pressure, $-\oint p\, n\, ds$, over the body surface found by the program; Eq. (8) is used only for comparison. C_p is shown only when a reference speed exists: V_∞ of the uniform flow, or a reference speed set by the user when the scene has no free stream.

3. PROGRAM ARCHITECTURE

3.1 Structure

The program is written in Python with PySide6 (Qt) for the interface, pyqtgraph for the plot, NumPy for the arrays and Numba for compiled velocity kernels. The code is divided into the layers given in Table 2. No interface code is included in the physics layer, so it is also used by the automatic tests and by the verification report. A scene can also be exported or verified from the command line without a window.

Table 2. Layers of the program

Layer	Content
physics	Elementary flows, superposition, method of images, scene checks, compiled kernels, streamlines, stagnation points, bodies, surface pressure and forces, verification
model	Scene model with undo / redo; scene file (.pflow) with elements, walls and plot settings
app	Main window, element palette, parameter dialogs, menus, plot layers, export, help
tests	Automatic tests of the physics and of the interface (pytest)

3.2 Computation of the Streamlines

After every add, edit or delete operation, the following steps are carried out. First, the scene is checked. Invalid parameters (for example $m = 0$), elements behind a wall and wedge angles that cannot be imaged are reported, and the message is shown in the parameter dialog before Apply is pressed. Coincident singular points are allowed, and only a note is shown (for

example the doublet and the vortex at the center of a lifting cylinder). Second, the image system of Section 2.3 is built. Third, evenly spaced streamlines are drawn with the method of Jobard and Lefer [4]: each line is integrated in the total velocity field with the fourth-order Runge–Kutta method, and a new line is started only where there is space for it. The spacing follows the streamline density set by the user. Direction arrows are placed along the lines in the direction of the velocity.

The velocity is never evaluated at the center of a source, vortex or doublet. A line is stopped at a small distance from such a point, and the point itself is not plotted. Stagnation points are found by Newton's method on $w(z) = 0$, and the dividing streamlines are started from them.

3.3 User Interface

The main window is shown in Figure 1. One icon is provided for each flow and wall type in the palette on the left, and its name is shown when the mouse is held over it. With a click, the compact parameter dialog with Apply and Cancel is opened (Figure 2(a)); the complex potential of the element is also shown in it. Labeled x and y axes are drawn in the plot in the middle. All current elements are listed with their parameters in the Elements menu (Figure 2(b)) and in the Elements panel, and a selected element can be edited or deleted at any time. New, Open, Save, Save as and Clear all are provided in the File menu (Figure 2(c)). The element types, the parameters, the walls and the plot settings are stored in a saved scene file. The plot limits (view range) and the streamline density are set in the plot settings panel.

The optional extensions of the assignment are also included: equipotential lines, velocity magnitude and C_p maps, zoom and pan, selectable and draggable element markers, and the export of figures (PNG, SVG, PDF) and data (CSV). Undo and redo, ready-made scenes, animated particles and a built-in user guide are also available.

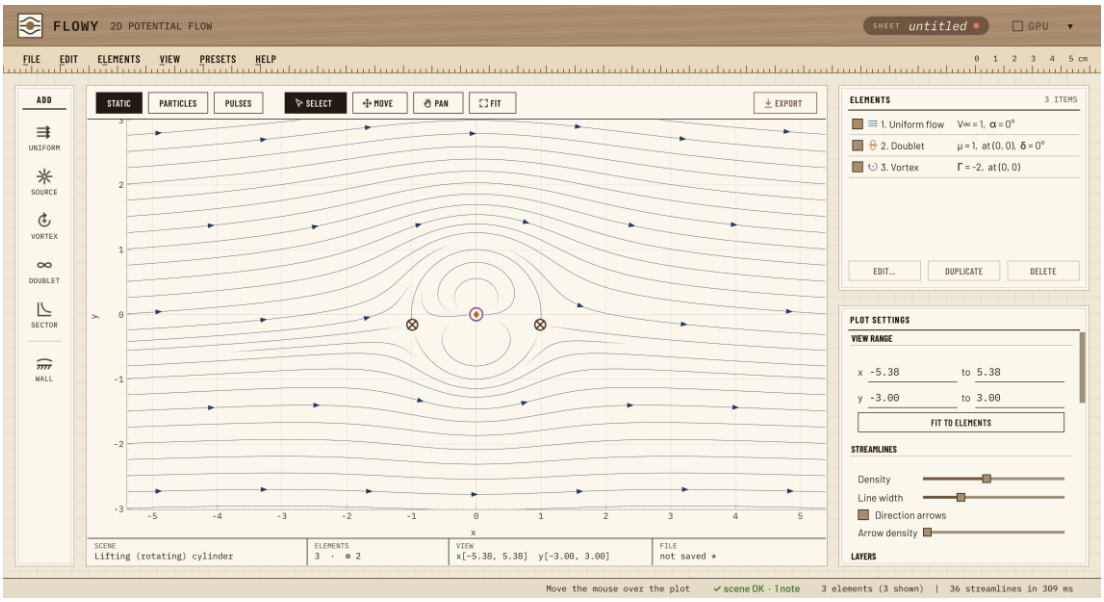
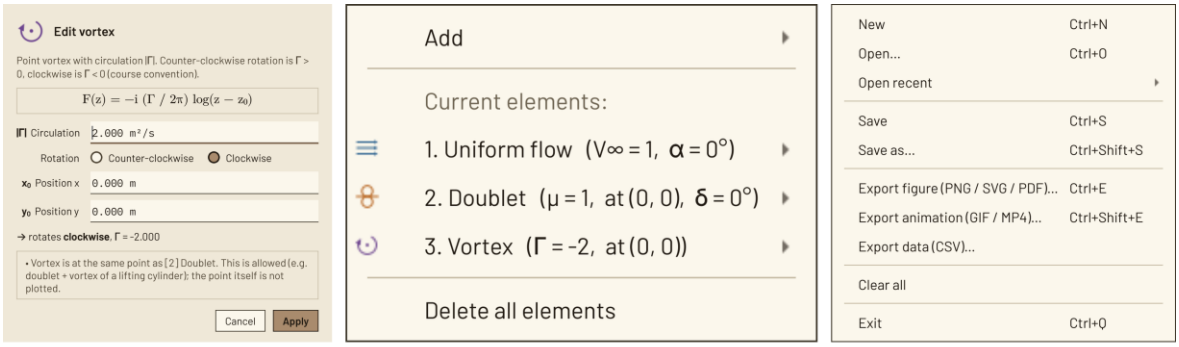


Figure 1. Main window: element palette (left), plot with streamlines and direction arrows (middle), Elements panel and plot settings (right); lifting cylinder scene



(a) (b) (c)

Figure 2. (a) Parameter dialog of a point vortex with Apply and Cancel, (b) Elements menu, (c) File menu

4. VALIDATION

A built-in verification is included in the program (Help → Run verification, or the command line). In total, 108 checks are carried out: 34 compare the velocity formulas with closed-form solutions at sample points, 57 check what the program draws and computes for the verification scenes, and 17 check the accuracy of the numerical methods (stream function along the drawn lines, stagnation point search, traced body shapes and lift). In each check, a program value is compared with an analytic value, and the check is passed when the error is below a stated tolerance. All 108 checks are passed. The results for the minimum verification cases of the assignment are given in Table 3, and the scenes are shown in Figure 3.

In Figure 3(a)–(c), the streamlines are straight lines at 45°, radial lines and circles, as expected. The outflow of the source and the counter-clockwise rotation of the vortex are shown by the arrows; for the sink and for $\Gamma < 0$, the directions are reversed. In Figure 3(d), the

stagnation points lie below the horizontal axis because of the circulation. In Figure 3(e) and (f), the walls are streamlines. The errors near 10^{-16} are round-off errors; the larger errors are caused by the numerical integration and by the traced body surface, and they remain below the tolerances. For case 5 of the assignment, every scene is saved to a file and opened again; the element list and the streamlines are identical.

Table 3. Results of the minimum verification cases

Case	Quantity checked	Error	Tolerance
Uniform, $\alpha = 0°, 45°$	Largest angle of the drawn lines to α	4.3×10^{-13} ∘	10^{-6} ∘
Source / sink $m = \pm 2$	Largest angle to the radial direction; sign of V_r	4.7×10^{-13} ∘	10^{-6} ∘
Vortex $\Gamma = \pm 2$	Circles: $(r_{max} - r_{min})/r$; sense of rotation	8.6×10^{-6}	10^{-4}
Cylinder $\mu = 1$	Surface $\lvert r - a \rvert$ [m]; C_p on it	0.0008; 0.0011	0.002; 0.005
Lifting, $\Gamma = -2$	Stagnation points [m]; L' [N/m]	10^{-16}; 2×10^{-4}	10^{-8}; 0.0025
Wall + source $m = 2$	Largest $\lvert V \cdot n \rvert$ on the wall [m/s]	0	10^{-9}
Sector flow, $\beta = 90°$	Largest $\lvert V \cdot n \rvert / \lvert V \rvert$ on both walls	1.2×10^{-16}	10^{-12}
Save → reopen (all)	Elements that differ; line shift [m]	0; 0	0.5; 10^{-12}

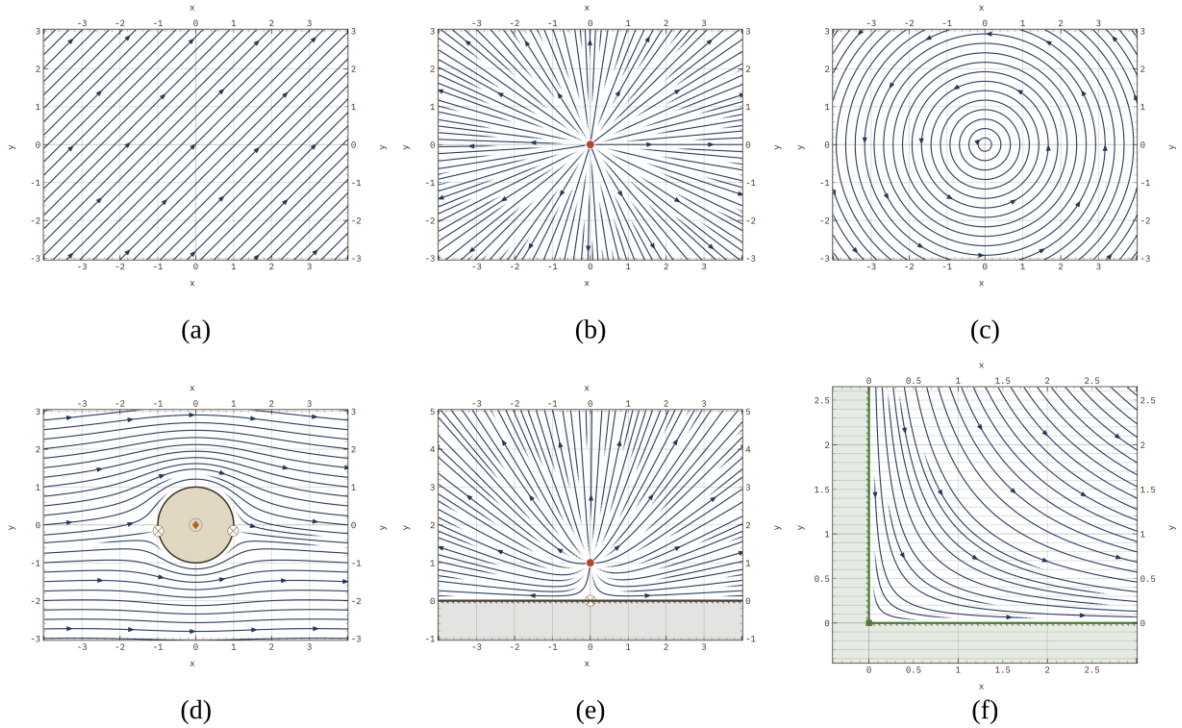


Figure 3. Verification scenes: (a) uniform flow, $\alpha = 45°$; (b) source, $m = 2$; (c) vortex, $\Gamma = 2$; (d) lifting cylinder, $\mu = 1$, $\Gamma = -2$; (e) source $m = 2$ at $(0,1)$ above a wall on the x axis; (f) sector flow, $\theta_1 = 0°$, $\theta_2 = 90°$

5. EXAMPLES

5.1 Lifting Cylinder

A uniform flow ($V_\infty = 1$ m/s, $\alpha = 0°$), a doublet ($\mu = 1$ m^3/s) and a clockwise vortex ($\Gamma = -2$ m^2/s) are placed at the origin, with $\rho_\infty = 1.225$ kg/m^3. The radius ([1], Eq. (3.98)), the stagnation points (Eq. (7)), the lowest C_p at $\theta = 90°$ (Eqs. (6) and (7)) and the lift (Eq. (8)) are:

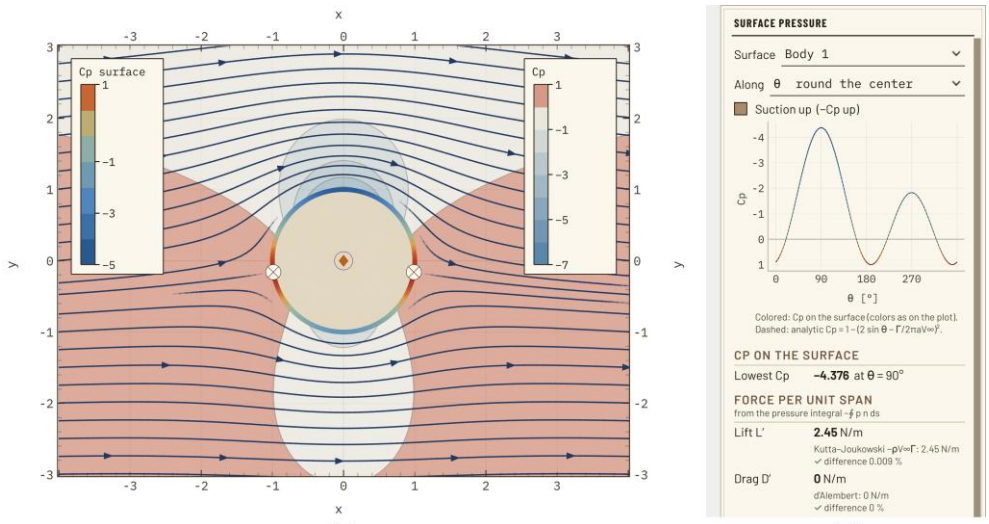
$$a = \sqrt{\frac{\mu}{V_\infty}} = \sqrt{\frac{1\ \text{m}^3/\text{s}}{1\ \text{m/s}}} = 1\ \text{m} \tag{9}$$

$$\sin\theta_s = \frac{\Gamma}{4\pi V_\infty a} = \frac{-2}{4\pi(1)(1)} = -0.1592 \ \Rightarrow\ \theta_s = -9.16°,\ 189.16° \tag{10}$$

$$C_p = 1 - \left(2\sin\theta - \frac{\Gamma}{2\pi a V_\infty}\right)^2 = 1 - \left(2 + \frac{2}{2\pi}\right)^2 = -4.375 \tag{11}$$

$$L' = -\rho_\infty V_\infty \Gamma = -(1.225)(1)(-2) = 2.45\ \text{N/m} \tag{12}$$

The program result is shown in Figure 4. Both stagnation points lie below the horizontal axis, and the flow over the top is faster. The computed and the analytic C_p curves coincide in Figure 4(b).



(a) (b)

Figure 4. Lifting cylinder ($V_\infty = 1$, $\mu = 1$, $\Gamma = -2$): (a) C_p map, body and C_p on the surface; (b) Surface pressure panel: C_p against θ (computed and analytic), lift and drag

5.2 Other Bodies

In the examples of Figure 5, $V_\infty = 1$ m/s is used. The nose of the Rankine half-body with $m = 2$ m^2/s ([1], Eqs. (3.78) and (3.79)) and the stagnation points of the Rankine oval with $m = 3$ m^2/s and the source and sink at $x = \mp b$, $b = 1$ m ([1], Eq. (3.81)) are:

$$x_{nose} = -\frac{m}{2\pi V_\infty} = -\frac{2}{2\pi(1)} = -0.3183\ \text{m} \tag{13}$$

$$x_s = \pm b\sqrt{1 + \frac{m}{\pi b V_\infty}} = \pm(1)\sqrt{1 + \frac{3}{\pi(1)(1)}} = \pm 1.398 \text{ m} \tag{14}$$

A cylinder above a wall and a source in a 90° corner are shown in Figure 5(c) and (d); the walls are streamlines because of the images. The main results are collected in Table 4.

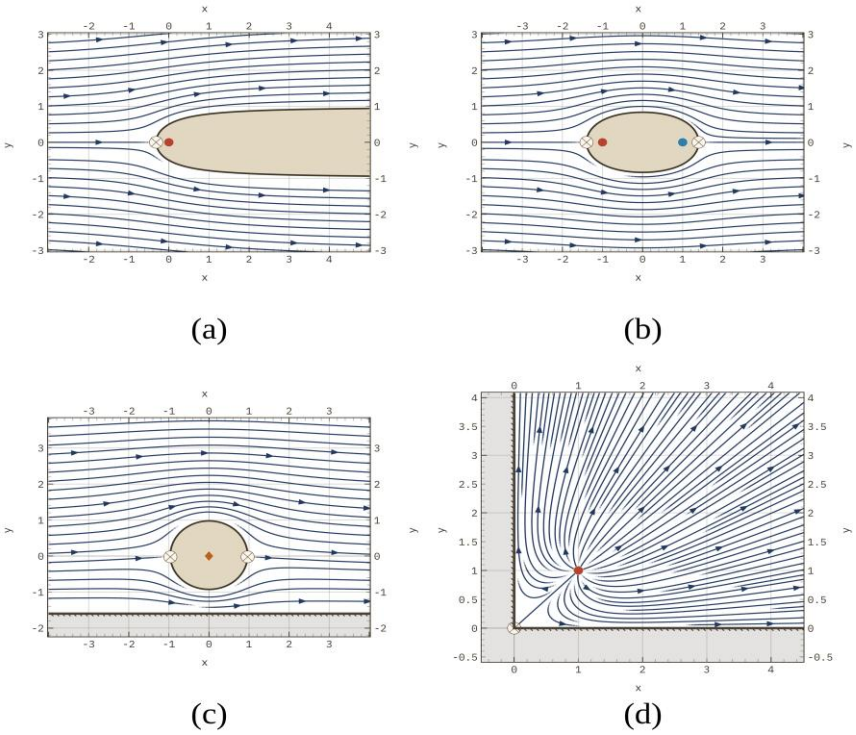


Figure 5. Examples: (a) Rankine half-body, $m = 2$; (b) Rankine oval, $m = \pm 3$ at $x = \mp 1$; (c) cylinder, $\mu = 1$, above a wall at $y = -1.6$; (d) source $m = 2$ at $(1,1)$ in a 90° corner of two walls

Table 4. Summary of the main results: hand calculation and program

Quantity	Equation	Hand calculation	Program
Cylinder radius a [m]	(9)	1	1.0008 (traced)
Stagnation angles θ_s [°]	(10)	-9.16; 189.16	-9.16; 189.16
Lowest C_p (at 90°)	(11)	-4.375	-4.376
Lift L' [N/m]	(12)	2.45	2.4502
Drag D' [N/m]	(8)	0	-4.2×10^{-6}
Half-body nose x_{nose} [m]	(13)	-0.3183	-0.3183
Oval stagnation points x_s [m]	(14)	± 1.398	± 1.398

6. CONCLUSION

A graphical program was developed in which the six elementary flows are superposed, the walls are enforced with images, the domain of the sector flow is checked, and the streamlines with direction arrows are redrawn after every change. All 108 verification checks are passed, and the examples agree with the hand calculations. Since an inviscid, incompressible, two-dimensional model is used, viscous drag and flow separation cannot be predicted.

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